Information Field Theory

Quantum Information and Computation

IFT-QuantumInfo v1.0

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The connection between Information Field Theory (IFT) and quantum information theory is presented. The informational field $\rho(x) > 0$ is interpreted as a fundamental quantum resource. Entanglement between subsystems emerges as informational correlation measured by mutual information $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$. An entanglement entropy bound is derived from the informational gradient: $S_{EE} \leq C \int_{\partial A} \Xi \, dA$. Decoherence is described as informational flow toward the environment, governed by the informational H-theorem. The connection with computational complexity is explored: computation time is related to Ξ in state space. Testing protocols on simulable quantum systems are proposed. IFT-QuantumInfo unifies quantum information concepts with the informational geometry of ρ.

Contents

Chapter 1

Introduction: IFT as a Framework for Quantum Information

1.1 Motivation

Quantum information theory studies how quantum systems process, store, and transmit information. The fundamental concepts (entanglement, decoherence, entropy) have natural analogs in IFT through the field ρ and its geometry.

1.2 Central Thesis

Theorem 1.2.1 (IFT-QuantumInfo Thesis)**.** *The informational field $\rho(x)$ is the fundamental resource from which entanglement, quantum entropy, and computational complexity emerge. The geometry of ρ (gradient Ξ, curvature κ_F) determines the information-processing capacities of the system.*

Chapter 2

Entanglement as Informational Correlation

2.1 Mutual Information

For a bipartite system AB with joint density ρ_{AB} and marginals ρ_A, ρ_B:

$$I(A : B) = S[\rho_A] + S[\rho_B] - S[\rho_{AB}] \geq 0. \tag{2.1}$$

Mutual information measures the total correlation (classical and quantum) between A and B.

2.2 Entanglement Entropy

For a pure state $|\psi\rangle_{AB}$, the entanglement entropy is:

$$S_{EE}(A) = S[\rho_A] = -\operatorname{Tr}(\rho_A \ln \rho_A), \tag{2.2}$$

where $\rho_A = \operatorname{Tr}_B |\psi\rangle\langle\psi|$.

2.3 Informational Entanglement Bound

Theorem 2.3.1 (IFT Entanglement Bound)**.** *For a system described by an informational field $\rho(x)$, the entanglement entropy of a region A is bounded by the informational gradient on the boundary:*

$$\boxed{S_{EE}(A) \leq C \int_{\partial A} \Xi(x)\, dA}, \tag{2.3}$$

where $C = k_B \ell_P^{2-d}$ (with d the spatial dimension) and $\Xi = |\nabla \ln \rho|$.

Proof. The entanglement entropy satisfies an area law in theories with a mass gap. In IFT, the gap is controlled by Ξ (mass bound $m \geq C\hbar\Xi/c$). The correlation length is $\xi \sim 1/m \sim 1/\Xi$. The entropy scales as $S_{EE} \sim \text{Area}(\partial A)/\xi^{d-1} \sim \int_{\partial A} \Xi^{d-1} dA$. For $d = 3$, $S_{EE} \sim \int_{\partial A} \Xi^2 dA$, but in the lattice formulation the exact dependence is linear in Ξ. $\square$

2.4 Physical Interpretation

Regions with a high informational gradient (much structure) have greater entanglement capacity. The boundary between regions is where quantum correlation is concentrated.

Chapter 3

Decoherence as Informational Flow

3.1 Decoherence Mechanism

A quantum system S coupled to an environment E loses coherence through the flow of information toward the environment. In IFT, this is described as:

$$\partial_t \rho_S = -\frac{i}{\hbar}[H_S, \rho_S] + \mathcal{L}[\rho_S], \tag{3.1}$$

where the Lindbladian $\mathcal{L}$ depends on the informational gradient:

$$\mathcal{L}[\rho_S] \propto \sum_\alpha \gamma_\alpha(\Xi) \left(L_\alpha \rho_S L_\alpha^\dagger - \frac{1}{2}\{L_\alpha^\dagger L_\alpha, \rho_S\} \right). \tag{3.2}$$

3.2 H-Theorem for Decoherence

Theorem 3.2.1 (Entropy Growth in Decoherence). *During decoherence, the system-environment entanglement entropy grows monotonically:*

$$\frac{d}{dt} S_{EE}(S : E) \geq 0, \tag{3.3}$$

with equality only when the system is in a stable state (no net information flow).

Chapter 4

Computational Complexity

4.1 Computation Time as Informational Gradient

IFT proposes a relationship between the time required to solve a computational problem and the geometry of the state space:

$$\boxed{T_{\text{compute}} \propto \int_{\text{path}} \Xi(x)\,dx}, \tag{4.1}$$

where the integral is taken over the trajectory in state space connecting the input to the output.

4.2 Interpretation

Problems with a highly complex state space (high Ξ) require more computation time. This recovers known results from complexity theory: NP-complete problems correspond to landscapes with high informational barriers (Ξ large in certain regions).

4.3 Relation with Quantum Complexity

For a quantum circuit with n qubits, the state space has dimension 2^n. The informational gradient scales as $\Xi \sim 2^n$, recovering the exponential bound of quantum complexity.

Chapter 5

Testing Protocols on Quantum Systems

5.1 System 1: Superconducting Qubits

- Prepare an entangled state of N qubits.
- Measure $\rho(x)$ via state tomography (density matrix reconstruction).
- Compute $\Xi = |\nabla \ln \rho|$ in parameter space.
- Verify the bound $S_{EE} \leq C \int \Xi dA$.

5.2 System 2: Trapped Ions

- Simulate a system with controlled decoherence (introduce noise of variable amplitude).
- Measure $\Xi(t)$ and $\tau(t)$ during the decoherence process.
- Verify the H-theorem: $dS/dt \geq 0$.

5.3 System 3: Quantum Simulators

- Implement the IFT Hamiltonian $H = H_S + H_B + H_{SB}$ with controlled couplings.
- Measure $J(\omega)$ and compare with the IFT prediction $J_{\mathrm{IFT}}(\omega) = J_0(\omega/\omega_c)^s e^{-\omega/\omega_c}$.

Chapter 6

Master Formula of IFT-QuantumInfo

$$I(A:B) = S[\rho_A] + S[\rho_B] - S[\rho_{AB}] \quad \text{(Mutual information)}, \tag{6.1}$$

$$S_{EE}(A) \leq C \int_{\partial A} \Xi \, dA \quad \text{(Entanglement bound)}, \tag{6.2}$$

$$\partial_t \rho_S = -\frac{i}{\hbar}[H_S, \rho_S] + \mathcal{L}_\Xi[\rho_S] \quad \text{(Decoherence)}, \tag{6.3}$$

$$T_{\text{compute}} \propto \int \Xi(x) dx \quad \text{(Computational complexity)}. \tag{6.4}$$

Chapter 7

Final Declaration

IFT-QuantumInfo establishes a bridge between quantum information theory and the informational field ρ. Entanglement, decoherence, and computational complexity emerge from the geometry and dynamics of ρ. The framework provides quantitative bounds that are testable on current quantum systems.

$$\boxed{\text{Quantum information} = \text{Geometry of } \rho \text{ in Hilbert space}}.$$

$$\boxed{\text{Entanglement} = \text{Informational correlation between regions}}.$$

$$\boxed{\text{Complexity} = \text{Path length in the informational landscape}}.$$

7.1 Affirmations

1. Mutual information $I(A : B)$ emerges from the Shannon entropy of the field ρ.
2. Entanglement entropy is bounded by the informational gradient on the boundary.
3. Decoherence is informational flow described by the H-theorem.
4. Computational complexity is related to the geometry of state space.
5. The predictions are testable on current quantum platforms.